

Evaluation of Salivary pH and buffering capacity in patients with Oral lichen planus:A Case-Control Study

Abstract

Oral Lichen Planus (OLP) is a common immune-mediated inflammatory disorder affecting the oral mucosa. Saliva maintains and protects the oral health by various mechanisms. Saliva can thus be used as a diagnostic tool for various oral pathologies including Oral lichen planus.

Aim: to determine the salivary pH and buffering capacity in oral lichen planus patient compare to healthy group.

Method: A case control study including 60 participants. Salivary pH was analysed using salivary pH metre and buffering capacity using GC Saliva Check Buffer strips.

Result: The analysed data showed a significant lowering of salivary pH and buffering capacity in OLP patients as compared to healthy individuals.

Conclusion: Salivary parameters pH and buffering capacity are important in maintaining oral health and their alteration in OLP reflects compromised salivary function and oral equilibrium.

1. Introduction

Oral lichen planus (OLP) is a chronic inflammatory autoimmune disease of unknown etiology (1–4). Most common oral mucosal lesion affecting 0.5–2% of adult, slightly more common in woman than in man (5,6). OLP may present as reticular, erythematous, ulcerative, plaque-like, bullous and papular lesions (5,7,8). The lesions can be asymptomatic, bilaterally/ symmetrically anywhere in the oral cavity affecting predominantly the buccal mucosa, gingiva and tongue (5,7), causing symptoms including burning sensation, severe pain which interfere with speaking, eating and swallowing (9,10) and in some cases dry mouth (11).

Saliva is a complex fluid with large number of properties and functions that plays an crucial role in maintaining the health and function of the oral cavity (12–14). Saliva exhibits significant buffering capacity, enabling the neutralization of bacterial and cariogenic acids and the maintenance of oral pH within a physiological range of approximately 6.5–7.5 (15). Saliva can be used as a diagnostic aid as it is considered as the “mirror” of the oral cavity. (16)

Although there is alteration in the salivary parameters in patients with OLP, the evidence regarding pH and buffering capacity is limited. Thus the aim of this case-control study is to evaluate salivary pH and buffering capacity in patients with Oral Lichen Planus and to compare these parameters with those of healthy individuals. Saliva collection is a patient-friendly alternative to invasive medical procedures since it is simple and does not require specialized medical skills (13,17).

2. Methods and Materials

This is a Case-Control study including 60 individuals: 30 patients with clinically and histopathologically diagnosed Oral Lichen Planus and 30 age and sex-matched healthy individuals. The selected participants were of age range 25-55 years with no systemic diseases nor on any medication affecting salivation.

All the patients were explained about the procedure. Patients were instructed not to eat or drink for 2 hours prior to the examination for better data collecting standardization. Unstimulated saliva was collected from all the individuals under standardised conditions. The collected saliva analysed using

- Salivary pH meter for salivary pH
- GC Saliva Check Buffer strips for buffering capacity

3. Result

The study group had 30 participants, with 18 males and 12 females with a mean age of 42.1 ± 6.5 years and smoking prevalence of 20%. The control group also included 30 participants, consisting of 17 males and 13 females with a mean age of 41.3 ± 5.9 and smoking prevalence of 13.3%.

Table 1: Demographic Distribution of Study Participants

Variables	OLP Cases(n=30)	Controls(n=30)	p-value
Age(Mean $\pm$ SD)	42.1 $\pm$ 6.5	41.3 $\pm$ 5.9	0.65
Gender(M/F)	18/12	17/13	0.79
Smoking Status.	6 (20%).	4 (13.3%).	0.48

The demographic characteristics of study participants showed no statistically significant difference between case and control group with regards to age, gender distribution and smoking status (Table 1).

Table 2: Comparison of Salivary Parameters between Cases and Controls

Parameter.	OLP Cases(n=30).	Controls(n=30).	t-value.	p-value
Salivary pH (Mean $\pm$ SD)	6.2 $\pm$ 0.4.	6.8 $\pm$ 0.3	6.42	<0.001
Buffering Capacity(Score/10)	4.2 $\pm$ 0.9	7.1 $\pm$ 0.6	213.27	<0.001

Table 2 shows that both salivary pH and buffering capacity were lower in OLP cases as compared to controls and the difference was statistically significant ($p < 0.001$).

Table 3: Salivary pH and buffering capacity based on clinical types of OLP

OLP Type	n	Mean Salivary pH $\pm$ SD	Mean Buffering Capacity $\pm$ SD
Reticular.	12	6.4 $\pm$ 0.3	4.8 $\pm$ 0.6
Erosive.	10	6.0 $\pm$ 0.2	3.9 $\pm$ 0.8
Atrophic	8	5.9 $\pm$ 0.4	3.7 $\pm$ 0.9

p-value(ANOVA)	<0.01	<0.01
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Salivary pH and buffering capacity in three clinical subtypes of Oral lichen planus were compared in Table 3, showing that both the parameters were lowest in atrophic form of OLP. The difference among the 3 types was statistically significant as the p-value <0.01, ANOVA.

Table 4 indicates a strong positive correlation between salivary pH and buffering capacity among OLP patients.

Table 4: Correlation between Salivary pH and buffering capacity among OLP Patients

Correlation pair.	Pearsons's r	p-value
Salivary pH vs Buffering Score	0.71.	<0.001*

4. Discussion

The chemical and physical properties of saliva play an important role in maintaining the health and functions of the oral cavity. The salivary flow rate, hydrogen ion concentration (pH), buffering capacity, and microflora influence the balance of oral health (2). Oral pathologies lead to alteration in the salivary parameters which influences the overall oral health (5). Decreased salivary secretion with changes in salivary pH was reported earlier (15).

Mouth dryness is one of the most common oral conditions associated with OLP (5,18,19), that attributes to xerostomia, salivary gland hypofunction or both. Salivary gland hypofunction significantly affects salivary flow rate, which in turn compromises oral lubrication, buffering capacity, antimicrobial activity, and overall oral health. (16)

In the present study it clearly demonstrated a marked difference in the salivary pH and buffering capacity in OLP patient compared to healthy controls. The mean salivary pH (6.2 ± 0.4) was significantly lower as compared to the control group (6.8 ± 0.3), salivary buffering capacity was markedly reduced in OLP cases (4.2 ± 0.9 vs. 7.1 ± 0.6 , $p < 0.001$). These findings suggest that OLP is associated with alterations in salivary chemistry, which may contribute to a disturbed oral environment.

On analysing the salivary parameters in clinical sub types of OLP ,the atrophic form exhibited lower pH(5.9 ± 0.4) and buffering capacity (3.7 ± 0.9) suggesting that severity and clinical presentation among various subtypes are directly associated with the same.

The present study also shows a positive correlation between salivary pH and buffering capacity among OLP patients(pearson's $r=0.71$, $p < 0.001$) indicating that when saliva pH lowers the buffering capacity is also lowered.

5. Conclusion

The present study demonstrates a significant association of Oral lichen planus and salivary parameters specifically the buffering capacity and the pH. OLP patients exhibited a lower pH and buffering capacity compared to healthy individuals suggesting an altered salivary equilibrium. Furthermore a marked difference in various clinical subtypes was found with atrophic form showing more severity suggesting that OLP has great influence on the salivary function. Assessment of salivary parameters may therefore serve as a useful, non-invasive diagnostic and prognostic tool in the evaluation and management of OLP.

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